



Inclusive Design Challenge 2026
Innovation Competition: Inclusive Design
Challenge 2026
Continuing Education Center, King Mongkut's
University of Technology Thonburi
4 August 2026, Suralai Hall, ICONSIAM,
Bangkok

Round 1 Submission: Proposal / Concept Screening

1. Project Title

- Thai title: กระจกอัจฉริยะผู้ช่วยส่งเสริมการพึ่งพาตนเองและการมีงานทำ สำหรับผู้ที่มีความบกพร่องด้านความจำและการรู้คิด (ข้อเสนอ ปรับได้)
- English title: MindMirror — An Ambient Assistive Smart Mirror Supporting Independence and Employment for Adults with Memory and Cognitive Disability (working title)

2. Developer Information

- Team members: List all team members.
- Organization / institution: Add organization or institution.
- Email: Add the primary contact email.
- Phone: Add the primary contact phone number.
- Advisor, if any: Add advisor name and affiliation.

3. Innovation Concept and Background

The problem. A person cannot hold a job if they cannot reliably manage the ordinary structure of a day — taking medication on time, remembering an appointment, following the

steps of a task in order. For adults living with cognitive and memory disability, this everyday structure is precisely what is impaired. Yet in Thailand this population is largely locked out of work. Of roughly 855,000 working-age people with disability, only about **36.3%** are employed, and merely **8%** participate in the competitive labour market [3, 4]. Cognitive-type disability — intellectual disability (about 145,000 registered) and mental or behavioural disability (about 167,000 registered) — makes up a large and especially under-served share of this group [2].

This exclusion is caused by the disability, not incidental to it. Across traumatic brain injury, post-stroke impairment, and intellectual disability, deficits in memory, attention, and executive function are the strongest predictors of whether a person can return to and keep a job [5, 7, 8] — after traumatic brain injury, unemployment can reach 60% two years post-injury [8]. The barrier to earning, for this group, is a barrier to remembering and organising.

Target group. Working-age adults (15–59) with cognitive or memory disability: intellectual disability, traumatic brain injury, post-stroke cognitive impairment, and early-onset dementia.

Why current solutions fall short. Thailand’s disability allowance (800 baht/month, roughly 29% of the poverty line) is income support, not a route to work [1]. The employment quota law is weakly enforced, and employers frequently pay fines rather than hire [6]. The assistive technology that does target cognition is almost entirely phone- or tablet-based — which fails the very users it is meant to help: it requires them to remember to open an app, to carry and charge a device, and to navigate menus. Abandonment is high when setup is complex or support is absent [14, 17, 18]. The memory aid should not itself demand memory to use.

Why an ambient smart mirror, and why now. A mirror is furniture people already pass every day: it needs no activation ritual, nothing to carry, and does not look like a medical device. It can offer support at the exact place and moment it is needed — the morning medication routine at the bathroom mirror, an orientation to the day, a step-by-step task prompt. This is not speculative: a smart-mirror cognitive-training system has already been built and shown to improve cognitive scores, recall, and orientation in adults with cognitive disability [9], smart mirrors for independent living have been demonstrated [10], and randomised evidence shows prompting technology reduces the assistance people with acquired brain injury need for daily activities [11]. The enabling technologies — reliable on-device voice, vision, and low-cost displays — have only recently matured enough to make an affordable, private, always-on mirror practical.

From Learning to Earning. Our thesis is a causal chain. Reliable daily routines make a person independent; independence makes them able to learn and re-learn work tasks; the ability to learn tasks makes them employable. The mirror is an ambient coach across that whole chain — it steadies the daily foundation and then carries the same step-by-step prompting into work skills. Assistive technology has been shown in systematic review to improve employment outcomes for people with cognitive disability [14]. We aim to turn daily independence into a genuine path to earning.

4. Project Objectives

1. To support independent daily living for adults with cognitive and memory disability through ambient, always-on prompting — medication reminders, temporal orientation, and step-by-step task guidance — with no device to carry and no app to remember to open.

2. To build from daily independence toward employability by extending the same prompting to work-related task sequences and routines, so that the innovation supports not only living but learning and earning.
3. To reduce the daily supervision burden on family caregivers, freeing their time and supporting their own ability to remain in paid work.
4. To deliver this through an affordable, non-stigmatising, privacy-by-design device suitable for real homes and, in future, supported workplaces in Thailand.

5. Target Users

Primary users. Working-age adults with cognitive or memory disability who live independently or semi-independently and want to build or keep their autonomy — and, where possible, employment.

- **Disability type / accessibility need:** intellectual disability; traumatic brain injury; post-stroke cognitive impairment; early-onset dementia. The shared, addressable need is impairment of memory, attention, and executive function [5, 7, 8].
- **Age range:** working-age adults, 15–59.
- **Usage context:** the home, positioned at a fixed point in the daily routine (bathroom or hallway); designed to extend into supported employment settings in a later phase.
- **Estimated number of target users:** of roughly 855,000 working-age people with disability in Thailand, the cognitive-disability segment (intellectual plus mental or behavioural, about 312,000 registered) represents the core addressable group, in the tens of thousands even under conservative reach assumptions [2, 3].

- **Secondary users:** family caregivers (who benefit from reduced supervision load) and, prospectively, job coaches and supported-employment providers.
- **Related organisations / communities:** the Department of Empowerment of Persons with Disabilities, disability-focused foundations and vocational-rehabilitation centres, community health services, and employers participating in the disability employment quota.

6. Innovation Details and Method

Explain how the innovation works. This section should make the concept concrete enough for reviewers to understand the design, technology, workflow, and prototype plan.

6.1 Design Approach

- Inclusive Design / Universal Design principles applied in the project.
- Accessibility Design decisions that support people with disabilities and other user groups.
- Key user needs, constraints, and assumptions.

6.2 Technology Used

- Hardware and software components.
- AI, IoT, mobile application, sensor, cloud, or other technical systems used.
- Development tools, platforms, datasets, or models.

6.3 System Workflow

Add diagrams or supporting figures where useful.

- System diagram.
- Flowchart.
- System architecture.
- Prototype images.

- User journey.

6.4 Innovation Strengths

Explain what makes the project different from existing solutions.

- Ease of use.
- Affordability.
- Support for multiple disability groups.
- AI-assisted analysis or personalization.
- Potential to increase employment or income opportunities.

Add tables, charts, prototype screenshots, or experimental results here if available.

7. Impact on People with Disabilities and Society

Reduced barriers in daily life. The mirror's core function is to make daily routines reliable without a caregiver present. Prompting technology has been shown in a randomised controlled trial to reduce the human support people with acquired brain injury need to complete activities of daily living [11], and scoping reviews confirm gains in independent completion of multistep tasks [12]. We target a measurable reduction in missed medication doses and in the time and assistance needed to complete a daily routine.

Increased access to work and income. The starting point is stark: only 36.3% of working-age people with disability in Thailand are employed, and just 8% reach the competitive labour market [3, 4]. Because assistive technology has been shown in systematic review to improve employment outcomes for people with cognitive disability [14], and prompting systems improve job task completion and accuracy [14, 15], the mirror's path from daily independence to work-task

prompting is a credible route to raising participation above that baseline.

Greater participation and dignity. A non-stigmatising device that looks like ordinary furniture — not a medical monitor — supports autonomy and social participation rather than signalling dependence. And by reducing the need for constant supervision, it frees family caregivers' time, supporting their own capacity to stay in or return to paid work.

Expected quantitative outcomes (pilot targets). As a Round-1 concept, these are measurable objectives to be validated in prototype testing, not results already obtained: a reduction in missed medication doses; a reduction in the time and assistance needed to complete a defined daily routine (task-time reduction is the clean, precedented metric [11, 15]); an increase in independent completion of trained work-task sequences; caregiver time reclaimed per week; and reach across the cognitive-disability segment of Thailand's approximately 855,000 working-age people with disability [3].

Honest note on the evidence. Current smart-mirror and prompting evidence is strong for short-term independence but thinner on long-term employment retention [13]. Our pilot is designed specifically to begin closing that gap — which is itself part of the project's contribution.

8. Feasibility for Real-World Use

Production and deployment. The device is assembled from commodity, off-the-shelf parts — a two-way mirror, a flat display, a single-board computer, a camera, and a microphone/speaker — which keeps the bill of materials low and avoids dependence on any custom hardware. Deployment is a single fixed installation in the home (bathroom or hallway), with no wearable to lose and no ongoing device management demanded of the user. Prior smart-mirror systems built from

comparable components demonstrate that the form factor is buildable today [9, 10].

Privacy and trust by design. Because the device includes a camera and microphone in the home of a cognitively vulnerable person, trust is a prerequisite for adoption, and privacy concerns are a documented barrier to assistive-technology uptake [16, 19]. Our design commitment is on-device processing of video and audio, minimal data retention, no cloud streaming of faces, and clear controls for the user and their family. This is a designed-in strength, not an afterthought.

Adoption strategy. Abandonment of assistive technology is driven by complexity, poor usability, and lack of support [17, 18]. We address these directly through co-design with users and caregivers, a deliberately simple interface that avoids deep menus, and a light training and support pathway — all identified in the literature as the factors that make adoption stick [17, 20].

Cost effectiveness and who pays. The strongest economic argument is avoided or delayed cost of care: every period a person remains safely independent, and every hour of caregiver supervision reclaimed, has a real value. Likely payers are family caregivers, disability foundations and vocational-rehabilitation programmes, and — through the employment quota and its associated funds — employers and public agencies, rather than the person with a disability directly.

Scaling, sustainability, and partnerships. A commodity-hardware, software-defined product scales through manufacturing and software updates rather than bespoke engineering. Partnerships give both a distribution route and the pilot data needed to validate impact and secure funding — public: the Department of Empowerment of Persons with Disabilities and community health services; private: employers under the disability quota and technology suppliers; community:

disability foundations, vocational-rehabilitation centres, and caregiver networks.

9. Conclusion

For adults with cognitive and memory disability, the barrier to earning is a barrier to remembering and organising the ordinary structure of a day — and in Thailand this leaves the great majority of them out of work. Existing aids fail because they demand the very capacity they are meant to replace: something to remember, to carry, to set up. Our innovation is an ambient smart mirror that removes those demands — always on, nothing to carry, looking like ordinary furniture — and provides medication reminders, orientation, and step-by-step prompting at the moment and place they are needed.

The design rests on evidence: smart-mirror cognitive support has been built and shown feasible, and prompting technology has randomised and systematic-review support for improving independence and employment-related outcomes. Our contribution is to unite that evidence in one affordable, privacy-respecting device that follows the chain deliberately *from learning to earning* — steadying daily independence, then carrying the same prompting into work skills. Built from commodity hardware, designed with its users, and aimed at a large under-served population, it is a practical step toward turning independence into income and dignity for people currently excluded from both.

10. References

1. Wattanadumrong B., Chaiboonsri C., Kadthan W. (2026). The Effects of Disability on Household Income and Poverty in Thailand: Evidence from the Socio-economic Survey. *Economies*, 14(5), 169. doi:10.3390/economies14050169
2. National Assembly of Disability Thailand (NADT) (2022). *Report on Disability Situation in Thailand 2565*. Department of Empowerment of Persons with Disabilities (DEP). <https://nadt.or.th/en/stat65.html>
3. National Statistical Office of Thailand & UNICEF Thailand (2023). *The 2022 Disability Survey*. Bangkok: NSO/UNICEF.
4. Cheausuwantavee T., et al. (2021). Employment for Persons with Disabilities in Thailand: Opportunities and Challenges in the Labor Market. *Journal of Population and*

- Social Studies*.
5. Wolf T.J., Doherty M., Boone A., Rios J., et al. (2019). Cognitive oriented strategy training augmented rehabilitation (COSTAR) for ischemic stroke: a pilot exploratory randomized controlled study. *Disability and Rehabilitation*, 43(2), 201-210. doi:10.1080/09638288.2019.1620877
 6. Government of Thailand (2007). *Persons with Disabilities Empowerment Act B.E. 2550*, Sections 33–35 (employment quota and concessions).
 7. Cunningham L.J., Stapleton T., Walsh K., Horgan F. (2026). Post-stroke cognitive impairment is associated with poorer return-to-work outcomes in working-aged stroke survivors: A systematic review. *Clinical Rehabilitation*. doi:10.1177/02692155261420770
 8. Cuthbert J.P., Harrison-Felix C., Corrigan J.D., Bell J.M., et al. (2015). Unemployment in the United States After Traumatic Brain Injury for Working-Age Individuals. *Journal of Head Trauma Rehabilitation*, 30(3), 160-174. doi:10.1097/HTR.000000000000090
 9. Kang B.R., Bae Y.H., Park S.H., Kang H.Y. (2025). Feasibility of a developed cognitive training system based on virtual reality with smart mirror for expert in community older cognitive disabled persons setting. *Journal of Alzheimer's Disease*, 105(2), 598-608. doi:10.1177/13872877251330147
 10. Chaparro J.D., Ruiz J.F.B., Romero M.J.S., Peño C.B., et al. (2021). The SHAPES Smart Mirror Approach for Independent Living, Healthy and Active Ageing. *Sensors*, 21(23), 7938. doi:10.3390/s21237938
 11. O'Neill B., Best C., O'Neill L., Ramos S.D.S., et al. (2018). Efficacy of a Micro-Prompting Technology in Reducing Support Needed by People With Severe Acquired Brain Injury in Activities of Daily Living: A Randomized Control Trial. *Journal of Head Trauma Rehabilitation*, 33(5), E33-E41. doi:10.1097/HTR.0000000000000358
 12. Stierle J., Ryan J.B., Katsiyannis A., Mims P., et al. (2023). Using smart phone technology to improve daily living skills for individuals with intellectual disabilities. *Journal of Applied Research in Intellectual Disabilities*, 36(5), 1169-1178. doi:10.1111/jar.13139
 13. Milhem-Midlej T., Szpiro S.F. (2025). Can video prompting be used to teach employment skills to older adults with moderate to moderate-severe intellectual developmental disabilities?. *Research in Developmental Disabilities*, 166, 105112. doi:10.1016/j.ridd.2025.105112
 14. Sauer A.L., Parks A., Heyn P.C. (2010). Assistive technology effects on the employment outcomes for people with cognitive disabilities: A systematic review. *Disability and Rehabilitation: Assistive Technology*, 5(6), 377-391. doi:10.3109/17483101003746360
 15. Mihailidis A., Melonis M., Keyfitz R., Lanning M., et al. (2015). A nonlinear contextually aware prompting system (N-CAPS) to assist workers with intellectual and developmental disabilities to perform factory assembly tasks: system overview and pilot testing. *Disability and Rehabilitation: Assistive Technology*, 11(7), 604-612. doi:10.3109/17483107.2015.1063713
 16. Yusif S., Soar J., Hafeez-Baig A. (2016). Older people, assistive technologies, and the barriers to adoption: A systematic review. *International Journal of Medical Informatics*, 94, 112-116. doi:10.1016/j.ijmedinf.2016.07.004
 17. Thordardottir B., Malmgren Fänge A., Lethin C., Rodriguez Gatta D., et al. (2019). Acceptance and Use of Innovative Assistive Technologies among People with Cognitive Impairment and Their Caregivers: A Systematic Review. *BioMed Research International*, 2019, 1-18. doi:10.1155/2019/9196729
 18. Randall K.N., Drew H., Gilman E.S., Dixon E. (2024). Assistive Technology Uses and Barriers in the Home and Workplace for Adults With Intellectual and Developmental Disabilities. *Journal of Applied Research in Intellectual Disabilities*, 38(1). doi:10.1111/jar.13306
 19. Pal D., Zhang X., Siyal S. (2021). Prohibitive factors to the acceptance of Internet of Things (IoT) technology in society: A smart-home context using a resistive modelling approach. *Technology in Society*, 66, 101683. doi:10.1016/j.techsoc.2021.101683
 20. Liu L., Stroulia E., Nikolaidis I., Miguel-Cruz A., et al. (2016). Smart homes and home health monitoring technologies for older adults: A systematic review. *International Journal of Medical Informatics*, 91, 44-59. doi:10.1016/j.ijmedinf.2016.04.007